

Notation reference

Thomas Serre · Computational Methods for Mind, Brain & Behavior · Fall 2026 · one notation for the whole semester

Symbols are introduced in the order the course meets them. Position and type face carry meaning: the same letter never means two things, and the same thing never has two letters.

Conventions

Object	Written	Example	Note
scalar	lowercase italic	a, x, y, c	a single number
vector	bold lowercase	$\mathbf{x}, \mathbf{w}, \mathbf{b}$	a column; $\mathbf{x}^\top$ is the row
component	italic, subscript	x_j, w_j	j th entry of $\mathbf{x}$
matrix	bold uppercase	$\mathbf{X}, \mathbf{W}, \mathbf{D}$	$\mathbf{X} \in \mathbb{R}^{N \times D}$, rows first
matrix entry	italic, two subscripts	x_{ij}, w_{jk}, d_{ik}	row, then column
tensor (3-D and up)	bold uppercase	$\mathbf{A}^{[l]}$	rank from context
set, space	blackboard / calligraphic	$\mathbb{R}^D, \mathcal{D}$	
object label	upright uppercase	A, B, C	an image, not a matrix

Theme 1 — Representational spaces

Indices.

Index	Notation	Example	Range
sample (image, stimulus)	superscript (i)	$\mathbf{x}^{(i)}$	$i = 1, \dots, N$
repetition (trial) of sample i	superscript (i, r)	$\mathbf{x}^{(i, r)}$	$r = 1, \dots, R$
component (pixel, voxel, unit)	subscript j	x_j	$j = 1, \dots, D$
matrix entry	subscripts ij	$x_{ij} = x_j^{(i)}$	row i : sample; column j : feature
RDM entry	subscripts ik	$d_{ik} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(k)})$	$\mathbf{D} \in \mathbb{R}^{N \times N}$
dissimilarity rule	lowercase subscript	$d_{\text{euc}}, d_{\text{cos}}, d_{\text{corr}}$	$1 - \cos \phi, 1 - r$

D is the number of measurements per object (the dimension of the space); N the number of objects.

Operations.

Operation	Written	Componentwise
transpose	$\mathbf{x}^\top, \mathbf{X}^\top$	column $\rightarrow$ row; $(\mathbf{X}^\top)_{ji} = x_{ij}$
sum, difference	$\mathbf{x} + \mathbf{y}, \mathbf{y} - \mathbf{x}$	entry by entry: $(\mathbf{y} - \mathbf{x})_j = y_j - x_j$
scaling	$c \mathbf{x}$	$c x_j$ for every j
mean over repetitions	$\mathbf{x}^{(i)} = \frac{1}{R} \sum_{r=1}^R \mathbf{x}^{(i,r)}$	one mean per component
dot product	$\mathbf{w}^\top \mathbf{x}$	$\sum_{j=1}^D w_j x_j = \ \mathbf{w}\ \ \mathbf{x}\ \cos \phi$
length (Euclidean norm)	$\ \mathbf{x}\ $	$\sqrt{\mathbf{x}^\top \mathbf{x}} = \sqrt{\sum_j x_j^2}$
unit direction	$\mathbf{w} / \ \mathbf{w}\ $	length 1
projection of $\mathbf{x}$ onto unit $\mathbf{w}$	$\mathbf{p} = (\mathbf{w}^\top \mathbf{x}) \mathbf{w}$	length $\mathbf{w}^\top \mathbf{x}$
matrix-vector product	$\mathbf{W} \mathbf{x}$	$(\mathbf{W} \mathbf{x})_j = \sum_k w_{jk} x_k$
stacking samples	$\mathbf{X}$, one sample per row	row i is $(\mathbf{x}^{(i)})^\top$
Euclidean distance	$d_{\text{euc}}(\mathbf{x}, \mathbf{y}) = \ \mathbf{y} - \mathbf{x}\ $	$\sqrt{\sum_j (y_j - x_j)^2}$
cosine similarity	$\cos \phi = \frac{\mathbf{x}^\top \mathbf{y}}{\ \mathbf{x}\ \ \mathbf{y}\ }$	$d_{\text{cos}} = 1 - \cos \phi$
correlation	$r = \frac{\mathbf{x}_c^\top \mathbf{y}_c}{\ \mathbf{x}_c\ \ \mathbf{y}_c\ }, \mathbf{x}_c = \mathbf{x} - \bar{x} \mathbf{1}$	$d_{\text{corr}} = 1 - r$
RDM	$d_{ik} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(k)})$	$\mathbf{D} \in \mathbb{R}^{N \times N}$, zero diagonal, symmetric

Theme 2 – Computing with a single neuron

Symbol	Name	Notes
$\mathbf{x}, \mathbf{y}, \hat{\mathbf{y}}$	input, target, prediction	bold lowercase column vectors
$\mathbf{w}, b$	incoming weights, bias	one neuron: $\mathbf{w}$ is a column vector
$z = \mathbf{w}^\top \mathbf{x} + b$	pre-activation	the one place a transpose appears
$a = g(z)$	activation	g generic; σ reserved for the sigmoid; ReLU, softmax by name
θ, η	all parameters, learning rate	θ collects every trainable weight and bias
$\mathcal{L}, \ell, \delta$	total loss, per-example loss, error	$\delta = \mathbf{y} - \hat{\mathbf{y}}$ (also the backprop delta)
$\mathcal{D}, \mathcal{B}, \mathbb{E}$	dataset, minibatch, expectation	$\mathcal{D} = \left\{ (\mathbf{x}^{(i)}, \mathbf{y}^{(i)}) \right\}_{i=1}^N$
$V, V_{\text{rest}}, V_{\text{th}}$	membrane, resting, threshold potential	threshold is V_{th} , not θ
I, τ_m, r	input current, time constant, firing rate	spiking and rate models
$s, f_i(s), s_i^{\text{pref}}$	stimulus, tuning curve, preferred stimulus	population codes
$\mathbf{r}, \hat{s}, I_F(s)$	population rate, decoded estimate, Fisher information	encoding and decoding

$$z = \mathbf{w}^\top \mathbf{x} + b, \quad a = g(z), \quad \hat{\mathbf{y}} \text{ uses } \sigma \text{ for probabilities.}$$

Theme 3 — Networks of neurons

Layer index. Superscript $[l]$, $l = 0, \dots, L$: $\mathbf{W}^{[l]}$, $\mathbf{b}^{[l]}$, $\mathbf{z}^{[l]}$, $\mathbf{a}^{[l]}$, with $\mathbf{a}^{[0]} = \mathbf{x}$ and $\mathbf{a}^{[L]} = \hat{\mathbf{y}}$. L is the number of layers, K the number of classes.

A layer has no transpose. Weights are stored output-rows-first, so $\mathbf{W}^{[l]} \in \mathbb{R}^{D_l \times D_{l-1}}$ (rows index the units of layer l , columns the units feeding in). Therefore

$$\mathbf{z}^{[l]} = \mathbf{W}^{[l]} \mathbf{a}^{[l-1]} + \mathbf{b}^{[l]}, \quad \mathbf{a}^{[l]} = g^{[l]}(\mathbf{z}^{[l]}), \quad \hat{\mathbf{y}} = \mathbf{a}^{[L]}.$$

This matches PyTorch `nn.Linear(in, out)`, whose weight has shape `(out, in)`. The transpose appears only in the single-neuron dot product $z = \mathbf{w}^\top \mathbf{x} + b$, where $\mathbf{w}$ is a column vector.

Theme 4 — Vision and convolutional networks

Channel index. Subscript c , $c = 1, \dots, C_l$: $A_{:,c}$ is one channel of the feature map $\mathbf{A}^{[l]} \in \mathbb{R}^{H_l \times W_l^{\text{sp}} \times C_l}$; $\mathbf{K}^{[l]}$ is the kernel bank.

$$\mathbf{A}^{[l]} = g(\mathbf{K}^{[l]} * \mathbf{A}^{[l-1]} + \mathbf{b}^{[l]}).$$

Theme 5 — Recurrence and dynamics

Time index. Superscript $\langle t \rangle$, $t = 1, \dots, T$: $\mathbf{h}^{\langle t \rangle}$, $\mathbf{x}^{\langle t \rangle}$, $\hat{\mathbf{y}}^{\langle t \rangle}$ are the state, input and output at time t ; T is the sequence length.

Cross weights use destination-from-source subscripts: $\mathbf{W}_{hx}$ maps $\mathbf{x} \rightarrow \mathbf{h}$, $\mathbf{W}_{hh}$ maps $\mathbf{h} \rightarrow \mathbf{h}$, $\mathbf{W}_{yh}$ maps $\mathbf{h} \rightarrow \hat{\mathbf{y}}$.

$$\mathbf{h}^{\langle t \rangle} = g(\mathbf{W}_{hh} \mathbf{h}^{\langle t-1 \rangle} + \mathbf{W}_{hx} \mathbf{x}^{\langle t \rangle} + \mathbf{b}_h), \quad \hat{\mathbf{y}}^{\langle t \rangle} = g(\mathbf{W}_{yh} \mathbf{h}^{\langle t \rangle} + \mathbf{b}_y).$$

Theme 6 — Attention and transformers

Tokens are rows. $\mathbf{X} \in \mathbb{R}^{T \times d_{\text{model}}}$ holds one token per row (the row-stacking convention of Theme 1). Queries, keys and values are $\mathbf{Q} = \mathbf{XW}_Q$, $\mathbf{K} = \mathbf{XW}_K$, $\mathbf{V} = \mathbf{XW}_V$; d_{model} , d_k , d_v are the model, key and value dimensions; head h is a superscript (h) , $h = 1, \dots, n_h$; $\mathbf{W}_O$ is the output projection.

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}.$$

Theme 7 – Generative models

Diffusion step. A subscript t , $t = 0, \dots, T$: $\mathbf{x}_0$ is the clean sample, $\mathbf{x}_t$ the noised sample, ϵ the noise, ϵ_θ the noise-prediction network, β_t the schedule, $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$. Sequence time is a superscript, diffusion time a subscript; they never co-occur.

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad \mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2].$$